

DevLens

AI-Powered Debugging & Root- Cause Analyzer

"Don't just find the error. Understand why it happened."

Domain: Developer Tools



Made with GAMMA

Debugging Is Still Largely Manual

Developers spend significant time searching codebases, tracing stack traces, and switching between logs, Git history, and API tools to locate the source of an error.

Context Switching

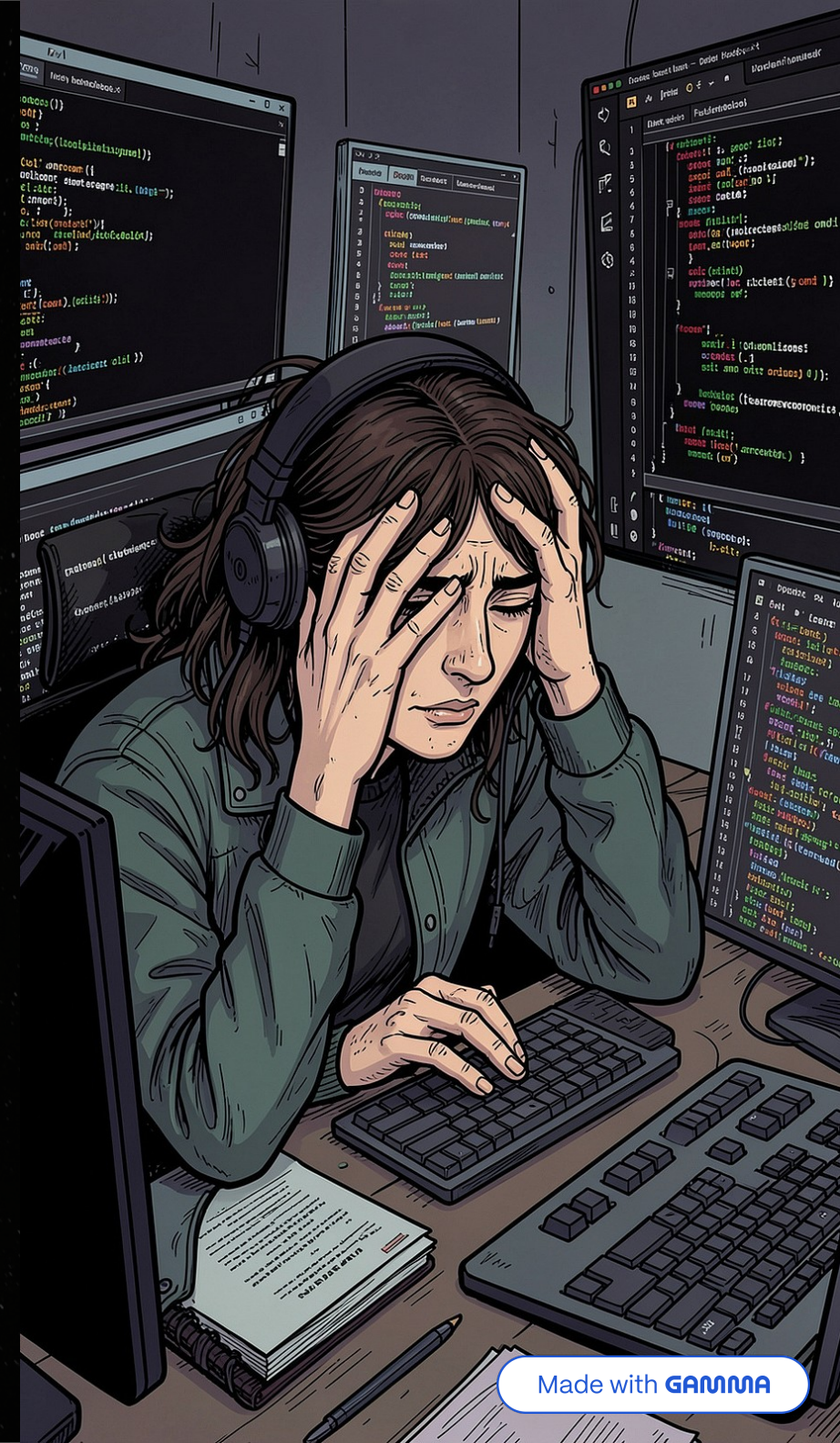
Jumping between logs, source code, Git history, and API tools fragments focus and slows resolution.

Missing Architecture Context

Existing AI assistants generate fixes without understanding the application's architecture, runtime evidence, or recent changes.

The Gap

Developers need an **investigator**, not just another chatbot — one that reasons over real evidence.



DevLens — An AI-Powered Debugging Investigator

DevLens analyzes a project as a **connected system**, not an isolated error. It combines multiple signals to produce structured, evidence-backed root-cause analysis.

What It Combines

- Source-code analysis & dependency mapping
- Stack-trace parsing & runtime logs
- API request information
- Git history & code changes
- AI-powered reasoning

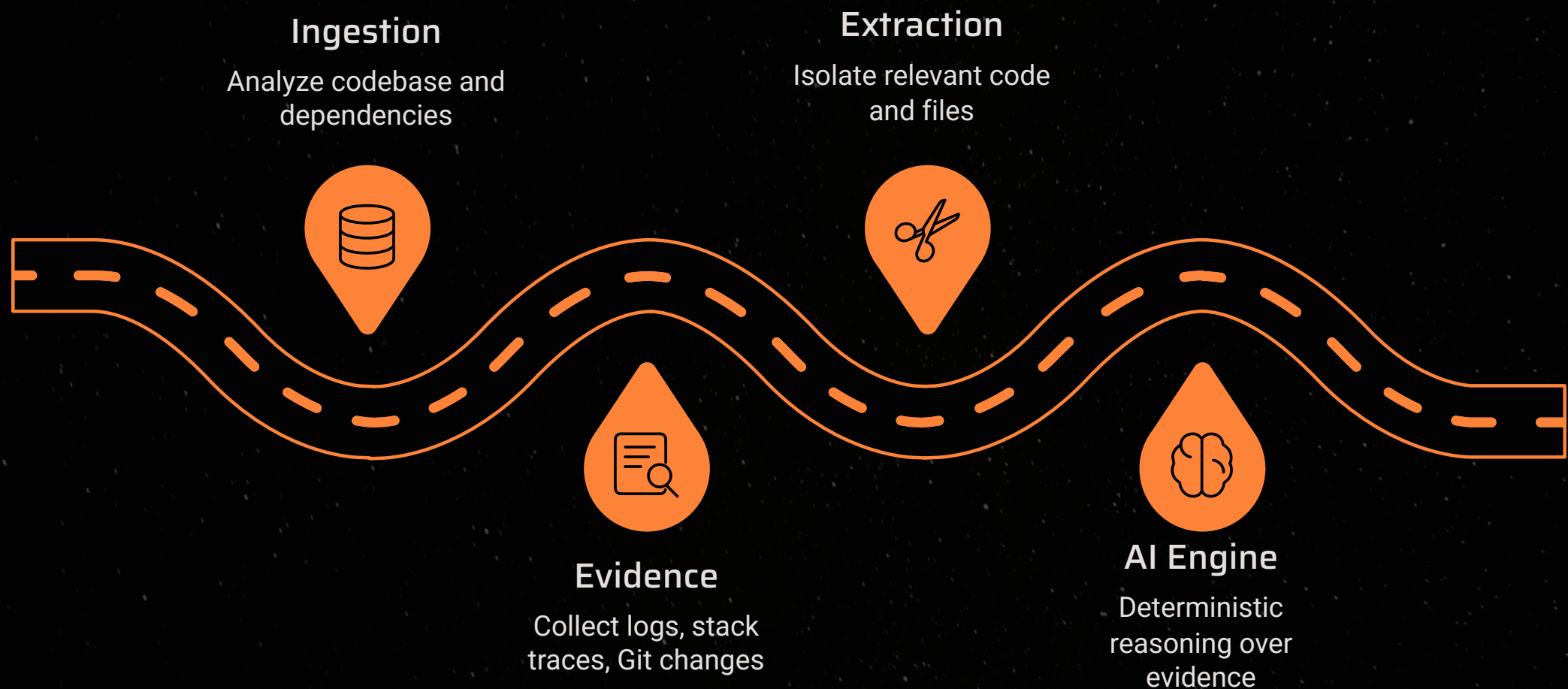
Output Structure

Every analysis produces:

- Likely root cause
- Supporting evidence
- Affected components
- Confidence score
- Suggested investigation or fix

From Error to Root Cause

DevLens follows a structured pipeline that narrows the search space before engaging the AI — sending only the **most relevant evidence** for analysis, not the entire repository.



□ Key principle: DevLens performs deterministic analysis first, then feeds a focused, structured context to the LLM — reducing noise and improving accuracy.

Beyond "Paste Your Error Into AI"

Traditional AI Debugging

Error → LLM → Suggested Fix

No context, no evidence, no confidence — just a guess based on raw text.

The DevLens Approach

**Error + Stack Trace + Code + Git + Logs →
Evidence Engine → AI Reasoning → Root
Cause + Confidence + Evidence**

Deterministic analysis and AI reasoning work together as layered components of one system.

📌 **AI is not the entire product** — it is one layer within a larger, evidence-driven debugging system.

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Example: API Returning HTTP 500

A developer receives POST /api/interview → 500 with `TypeError: Cannot read properties of undefined`. DevLens traces the request through the full stack.

 **Likely Root Cause**
interviewController.js:42 –
req.body.questions accessed
without validation.

 **Confidence: 87%**
Error at line 42, missing
questions field, no validation
middleware, recent schema
change.

 **Recommendation**
Inspect request validation and the recent schema modification.

What DevLens Provides



Codebase Intelligence

Understand files, functions, imports, routes, and dependencies across the entire project.



Error Analysis

Parse stack traces, runtime errors, and logs to pinpoint failure origins.



AI Root-Cause Analysis

Reason over structured evidence instead of raw error text.



Dependency & Call Graph

Visualize how components interact and trace the flow of execution.



Git-Aware Debugging

Identify recent changes that may have introduced regressions.



Visual Debugging Report

Root cause, evidence, affected components, and confidence in one place.

Technology Stack & Architecture

Stack

Frontend

React · Tailwind CSS ·
React Flow

Backend

Node.js · Express.js ·
REST APIs

Data

MongoDB

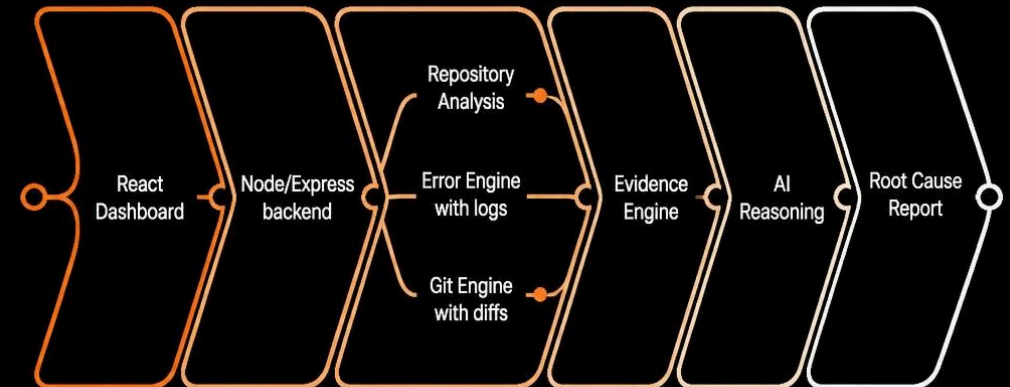
Code Intelligence

AST parsing ·
Dependency analysis · Git
integration · Stack-trace
parsing

AI Layer

LLM API · Structured JSON outputs · Context-aware
reasoning

Architecture Flow



MVP & Future Scope

Hackathon MVP — Phase 1

- JavaScript / TypeScript repository support
- Repository upload & codebase analysis
- Stack-trace parsing & error/log ingestion
- Relevant code extraction
- AI root-cause analysis
- Visual debugging report

Future Roadmap

01

Automated Bug Reproduction — Reproduce errors in a sandbox environment.

02

Regression Detection — Flag risky changes before deployment.

03

AI Test Generation — Generate tests for suspected failure scenarios.

04

Automated Fix Validation — Apply patches and verify via tests.

05

CI/CD Integration — Analyze failures automatically during builds.

06

Multi-Language Support — Python, Java, Go, C++, and more.



Why DevLens?

For Developers

- Reduce debugging time and context switching
- Understand unfamiliar codebases faster
- Identify root causes, not just symptoms

For Teams

- Faster incident resolution
- Better regression understanding
- Easier onboarding into existing codebases

Our Vision

Transform debugging from a manual search process into an **evidence-driven investigation**.

DevLens — Understand your code. Trace the failure. Find the root cause.